

Chapter 06 Part 2: Real Gases

1. A 1.0 L sample of a pure gas is found to have a lower pressure than that predicted by the ideal gas law. The best explanation for the observation is that the molecules of the gas
- (A) have a combined volume that is too large to be considered negligible when compared to the volume of the container
 - (B) have a low molecular mass and therefore do not strike the container walls with as much force as expected
 - (C) are attracted to each other and do not exert as much force on the container walls as they would if they had no mutual attractions ✓
 - (D) are attracted to the sides of the container and strike the container walls with more force than expected
2. A vessel contains Ar(g) at a high pressure. Which of the following statements best helps to explain why the measured pressure is significantly greater than the pressure calculated using the ideal gas law?
- (A) The molar mass of Ar is relatively large.
 - (B) A significant number of Ar₂ molecules form.
 - (C) The attractive forces among Ar atoms cause them to collide with the walls of the container with less force.
 - (D) The combined volume of the Ar atoms is too large to be negligible compared with the total volume of the container. ✓
3. When the actual gas volume is greater than the volume predicted by the ideal gas law, the explanation lies in the fact that the ideal gas law does NOT include a factor for molecular.
- (A) volume ✓
 - (B) mass
 - (C) velocity
 - (D) attractions
 - (E) shape
4. Ar(g) deviates more from ideal behavior at extremely high pressures than Ne(g) does. Which of the following is one reason for this difference?
- (A) The particle volume of Ar is greater than that of Ne. ✓
 - (B) Ar atoms have more valence electrons than Ne atoms have, so Ar atoms have greater interparticle forces.
 - (C) The intermolecular forces between Ne atoms are greater than those between Ar atoms.
 - (D) Ar atoms are more attracted to the walls of the container than Ne atoms are.
5. At constant temperature, the behavior of a sample of a real gas more closely approximates that of an ideal gas as its volume is increased because the

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- (A) collisions with the walls of the container become less frequent
- (B) average molecular speed decreases
- (C) molecules have expanded
- (D) average distance between molecules becomes greater ✓
- (E) average molecular kinetic energy decreases
6. Under which of the following conditions of temperature and pressure will H_2 gas be expected to behave most like an ideal gas?
- (A) 50 K and 0.10 atm
- (B) 50 K and 5.0 atm
- (C) 500 K and 0.10 atm ✓
- (D) 500 K and 50 atm
7. Under which of the following conditions of temperature and pressure would 1.0 mol of the real gas $\text{CO}_2(\text{g})$ behave most like an ideal gas?

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(A)	Temperature(K)	Pressure(atm)
	100	0.1

(B)	Temperature(K)	Pressure(atm)
	100	100

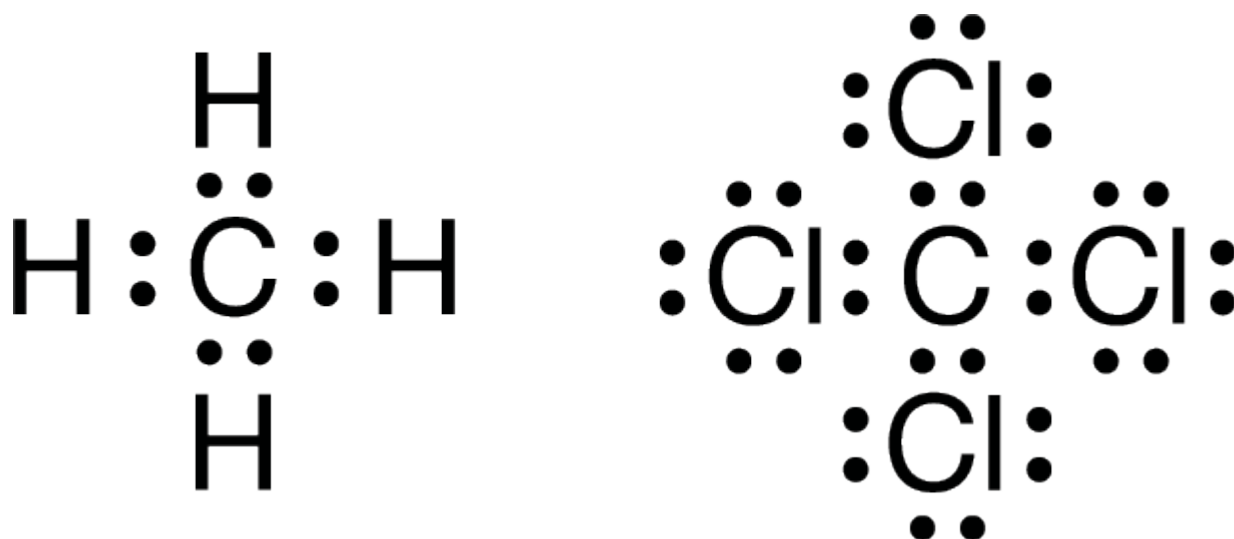
(C)	Temperature(K)	Pressure(atm)	✓
	800	0.1	

(D)	Temperature(K)	Pressure(atm)
	800	1

(E)	Temperature(K)	Pressure(atm)
	800	100

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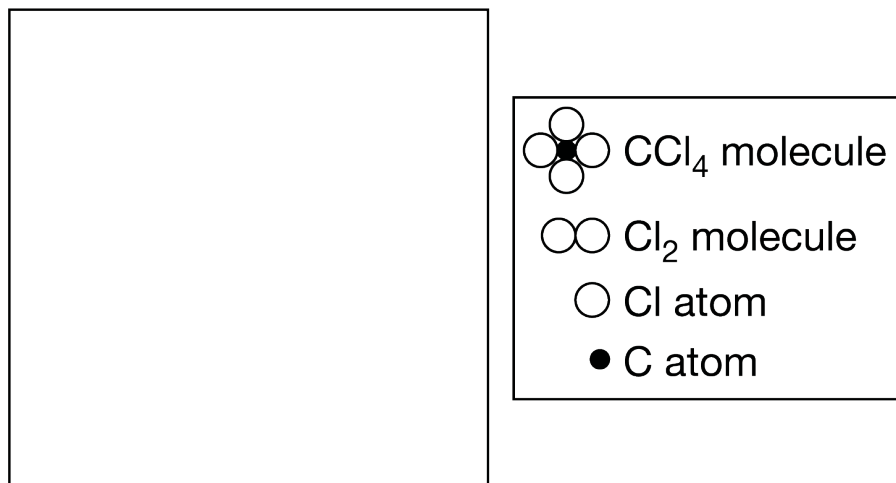
8. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Answer the following questions about CH_4 and CCl_4 . The Lewis electron-dot diagrams for molecules of the compounds are shown above.

- (a) Which compound has the higher boiling point? Justify your answer. The type(s) of intermolecular force(s) in both substances should be included in your answer.
- (b) At 400 K both compounds are gases. At this temperature, which compound, $\text{CH}_4(g)$ or $\text{CCl}_4(g)$, behaves more like an ideal gas? Justify your answer, including reasoning about both molecules.
- (c) $\text{CCl}_4(l)$ is placed in a previously evacuated container at 30°C , and some of the $\text{CCl}_4(l)$ evaporates. In the box below, draw a particulate diagram to show the species in the container after some of the $\text{CCl}_4(l)$ has evaporated. The phases of the species should be indicated by the spacing and distribution of the particles in the diagram. Some of the species shown in the legend will not be used.

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Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the criteria below:

- ☐ The response indicates that both compounds have London dispersion forces. (If the response indicates that either of the compounds has any other type of intermolecular force, the criterion is not met.)
- ☐ The response indicates that CCl₄ has the higher boiling point and that it has greater intermolecular forces.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response meets both of the criteria below:

- ☐ The response indicates that CH₄ behaves more like an ideal gas.
- ☐ The response indicates that CCl₄ has greater intermolecular forces, OR that the CCl₄ molecules are more attracted to each other, OR that CCl₄ has a larger electron cloud.

Part (c)

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The drawing meets all of the criteria below:

- ☐ There are closely spaced CCl_4 undefined molecules near the bottom of the box.
- ☐ There are more widely spaced CCl_4 undefined molecules in the box.
- ☐ There are no separate Cl_2 undefined molecules, Cl undefined atoms, or C undefined atoms in the box.

The following questions refer to the below.

The table below contains information about samples of four different gases at 273 K. The samples are in four identical rigid containers numbered 1 through 4.

Container	Gas	Pressure (atm)	Mass of Sample (g)
1	He	2.00	?
2	Ne	2.00	?
3	?	2.00	16.0
4	SO_2	1.96	64.1

9. The best explanation for the lower pressure in container 4 is that SO_2 molecules

- (A) have a larger average speed than the other three gases
- (B) occupy a larger portion of the container volume than the other three gases
- (C) have stronger intermolecular attractions than the other three gases
- (D) contain π bonds, while the other gases contain only σ bonds



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10. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is doing experiments with $\text{CO}_2(g)$. Originally, a sample of the gas is in a rigid container at 299 K and 0.70 atm. The student increases the temperature of the $\text{CO}_2(g)$ in the container to 425 K.

- (a) Describe the effect of raising the temperature on the motion of the $\text{CO}_2(g)$ molecules.
- (b) Calculate the pressure of the $\text{CO}_2(g)$ in the container at 425 K.
- (c) In terms of kinetic molecular theory, briefly explain why the pressure of the $\text{CO}_2(g)$ in the container changes as it is heated to 425 K.
- (d) The student measures the actual pressure of the $\text{CO}_2(g)$ in the container at 425 K and observes that it is less than the pressure predicted by the ideal gas law. Explain this observation.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the average speed of the molecules increases as temperature increases.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows a calculation similar to the following.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{0.70 \text{ atm}}{299 \text{ K}} = \frac{P_2}{425 \text{ K}} \Rightarrow P_2 = 0.99 \text{ atm}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response meets one or both of the following criteria.

- ☐ · The response indicates that faster-moving gas particles collide more frequently with the walls of the container, thus increasing the pressure.
- ☐ · The response indicates that faster-moving gas particles collide more forcefully with the walls of the container, thus increasing the pressure.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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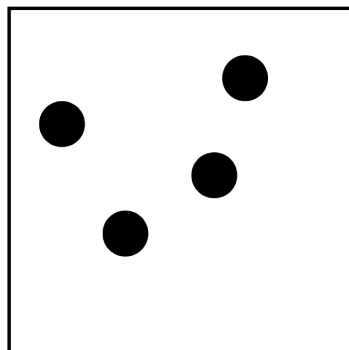
The response indicates that the attractive forces between CO_2 molecules result in a pressure that is lower than that predicted by the ideal gas law.

11. $\text{CCl}_4(g)$ deviates more from ideal gas behavior than $\text{CH}_4(g)$ does. Which of the following statements best explains this observation?

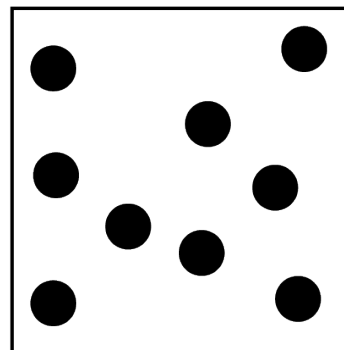
- (A) The mass of the CCl_4 molecule is greater than that of the CH_4 molecule.
- (B) The bond energy of the C-Cl bond is greater than that of the C-H bond.
- (C) The dipole-dipole forces between CCl_4 molecules are greater than those between CH_4 molecules.
- (D) The London dispersion forces between CCl_4 molecules are stronger than those between CH_4 molecules. ✓

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12.



Sample 1



Sample 2

The diagrams above represent two samples of Xe gas in containers of equal volume at 280 K. Which of the following correctly compares the two samples in terms of their deviation from ideal gas behavior and explains why?

- (A) The gas in sample 1 would deviate more from ideal behavior because the average distance an Xe atom travels before colliding with another Xe atom is greater.
- (B) The gas in sample 2 would deviate more from ideal behavior because the Xe atoms are closer together, leading to an increase in intermolecular attractions. ✓
- (C) The gas in sample 2 would deviate more from ideal behavior because the average speed of the Xe atoms is less, leading to an increase in intermolecular attractions.
- (D) The gases in both sample 1 and sample 2 would show the same deviation from ideal behavior.

13.

P (atm)	$V_{\text{experimental}}$ (L)	V_{ideal} (L)
0.200	122.2	122.2
2.000	12.17	12.22
8.000	2.925	3.055

The pressure of a sample of $\text{NH}_3(g)$ was changed by adjusting the volume of the container. The experimental volumes that were needed to achieve the desired pressures were different from the volumes predicted by the ideal gas law as shown in the table above. Which of the following best explains the difference between the experimental volumes and the volumes predicted by the ideal gas law?

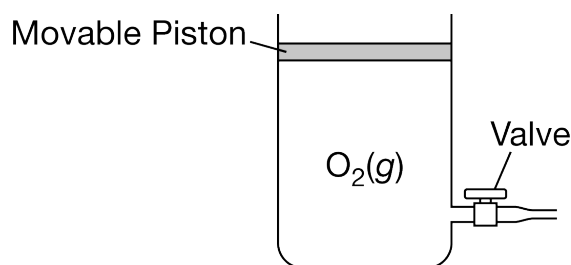
- (A) The attraction between the molecules becomes significant at high pressure. ✓
- (B) The repulsion between the molecules becomes significant at high pressure.
- (C) The average speed of the molecules decreases as the pressure increases.
- (D) The average speed of the molecules increases as the pressure increases.

14. Which of the following best helps explain why the pressure of a sample of $\text{CH}_4(g)$ (molar mass 16 g/mol) is closer to the pressure predicted by the ideal gas law than a sample of $\text{NH}_3(g)$ (molar mass 17 g/mol) ?

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- (A) NH_3 molecules are polar while CH_4 molecules are not, and the greater attractions between NH_3 molecules cause the molecules to collide with the walls of the container with less force. ✓
- (B) NH_3 molecules have a greater molar mass than CH_4 molecules, so the NH_3 molecules collide with the walls of the container with more force.
- (C) CH_4 molecules have more hydrogen atoms than NH_3 molecules, so CH_4 molecules have more hydrogen bonding and greater intermolecular forces.
- (D) CH_4 molecules are larger than NH_3 molecules, so the actual CH_4 molecules take up a significant portion of the volume of the gas.

15. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student investigates gas behavior using a rigid cylinder with a movable piston of negligible mass, as shown in the diagram above. The cylinder contains 0.325 mol of $\text{O}_2(g)$.

- (a) The cylinder has a volume of 7.95 L at 25°C and 1.00 atm. Calculate the density of the $\text{O}_2(g)$, in g/L, under these conditions.
- (b) Attempting to change the density of the $\text{O}_2(g)$, the student opens the valve on the side of the cylinder, pushes down on the piston to release some of the gas, and closes the valve again. The temperature of the gas remains constant at 25°C . Will this action change the density of the gas remaining in the cylinder? Justify your answer.
- (c) The student tries to change the density of the $\text{O}_2(g)$ by cooling the cylinder to -55°C , which causes the volume of the gas to decrease. Using principles of kinetic molecular theory, explain why the volume of the $\text{O}_2(g)$ decreases when the temperature decreases to -55°C .
- (d) The student further cools the cylinder to -180°C and observes that the measured volume of the $\text{O}_2(g)$ is substantially smaller than the volume that is calculated using the ideal gas law. Assume all equipment is functioning properly. Explain why the measured volume of the $\text{O}_2(g)$ is smaller than the calculated volume. (The boiling point of $\text{O}_2(l)$ is -183°C .)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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The response provides the correct calculated value.

$$0.325 \text{ mol O}_2 \times \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} = 10.4 \text{ g O}_2$$

$$D = \frac{m}{V} = \frac{10.4 \text{ g}}{7.95 \text{ L}} = 1.31 \text{ g/L}$$

OR

$$D = \frac{m}{V} = \frac{P(MM)}{RT} = \frac{(1.0 \text{ atm})(32.00 \text{ g/mol})}{(0.08206 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}})(298 \text{ K})} = 1.31 \text{ g/L}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response provides a correct answer and a valid justification.

No, the density of the gas remains constant because P , R , and T remain constant AND the mass and volume of O_2 decrease proportionately.

OR

A mathematical justification is shown below.

$$D = \frac{m}{V} = \frac{n \text{ moles of O}_2 \times \text{molar mass of O}_2}{\frac{nRT}{P}} = \frac{P \times (\text{molar mass of O}_2)}{RT}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response provides a valid explanation.

As the gas cools, the average kinetic energy (speed) of the O_2 molecules decreases. The molecules rebound with less energy when they collide with each other and the walls of the container. The spacing between particles decreases, causing

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the volume occupied by the gas to decrease.

OR

· As the gas cools, the average kinetic energy (speed) of the O_2 molecules decreases. The molecules rebound with less energy when they collide with each other and the walls of the container. The only way for the molecules to maintain a constant rate of collisions with the walls of the container (maintaining a pressure of 1.00 atm) is for the volume of the gas to decrease.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides a valid explanation.

· The ideal gas law assumes that gas particles do not experience interparticle attractions. As a real gas cools further, the intermolecular forces have greater effect as the average speed of the molecules decreases, resulting in inelastic collisions. To maintain a gas pressure of 1.00 atm, the volume must decrease to accommodate more collisions with less energy.

16. The ideal gas law best describes the properties of which of the following gases at 0°C and 1 atm?

- (A) PH_3
- (B) HBr
- (C) SO_2

(D) N_2



17. Which of the following behaves most like an ideal gas at the conditions indicated?

(A) $\text{H}_2(\text{g})$ molecules at 10^{-3} atm and 200°C

(B) $\text{O}_2(\text{g})$ molecules at 20 atm and 200°C

(C) $\text{SO}_2(\text{g})$ molecules at 20 atm and 200°C

(D) $\text{NH}_3(\text{g})$ molecules at 20 atm and 200°C

(E) $\text{NH}_3(\text{g})$ molecules at 20 atm and 300°C



18. Which of the following gases deviates most from ideal behavior?

(A) SO_2

(B) Ne

(C) CH_4

(D) N_2

(E) H_2

